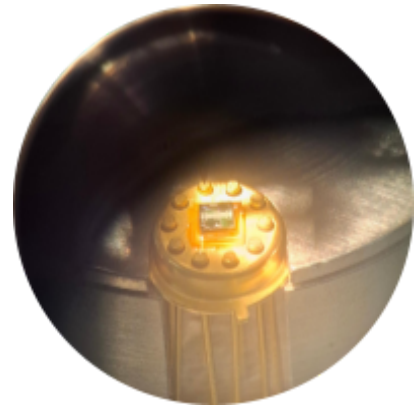


## Gas sensor based on tungsten trioxide ( $\text{WO}_3$ ) nanoparticles

### I. Features

- Low power consumption
- Easy-to-use
- Small size
- Low cost
- Detection of multiple gases:  $\text{N}_2/\text{O}_2$ , Ethanol ( $\text{CH}_3\text{CH}_2\text{OH}$ ), Methane ( $\text{CH}_4$ )
- Integrated temperature sensor (Aluminium resistor)
- Integrated heater (Polysilicon resistor)
- 



### II. Description

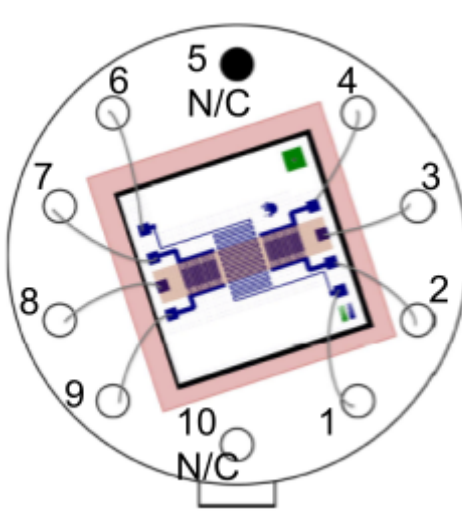
This gas sensor is designed for monitoring air quality and detecting specific gases. It is based on tungsten trioxide ( $\text{WO}_3$ ) nanoparticles. The sensor includes a heater and a temperature sensor, allowing measurement of the operating environment.

### III. Specifications

| Parameter                | Value/Description                                                    |
|--------------------------|----------------------------------------------------------------------|
| Type                     | Nanoparticle gas sensor (based on $\text{WO}_3$ nanoparticles)       |
| Sensing principle        | By resistance variation                                              |
| Materials                | Silicon, Aluminium, Polysilicon, Tungsten trioxide ( $\text{WO}_3$ ) |
| Power supply requirement | 12V for heating resistors, 5V for Gaz sensors resistors              |
| Nature of output signals | Analog signal (resistance variation)                                 |

|                             |                                                                                              |
|-----------------------------|----------------------------------------------------------------------------------------------|
| <b>Nature of measurands</b> | Resistance ( $G\Omega$ range)                                                                |
| <b>Package</b>              | 10-Lead TO-5 metal can and cap                                                               |
| <b>Diameter</b>             | 8mm                                                                                          |
| <b>Height</b>               | 21mm                                                                                         |
| <b>Mounting</b>             | Through-hole or surface-mount (depending on application)                                     |
| <b>Detectable gases</b>     | $N_2/O_2$ , Ethanol ( $CH_3CH_2OH$ ), Methane ( $CH_4$ )                                     |
| <b>Typical applications</b> | Air quality monitoring, IoT smart objects, Safety (leak detection), Environmental monitoring |

#### IV. Pins configuration

| Pin number | Usage                 |  |
|------------|-----------------------|--------------------------------------------------------------------------------------|
| 3          | Thermistor            |                                                                                      |
| 8          | Thermistor            |                                                                                      |
| 7          | Gas sensor resistor 1 |                                                                                      |
| 9          | Gas sensor resistor 1 |                                                                                      |
| 2          | Gas sensor resistor 2 |                                                                                      |
| 4          | Gas sensor resistor 2 |                                                                                      |
| 1          | Heating resistor      |                                                                                      |
| 6          | Heating resistor      |                                                                                      |
| 5, 10      | Not Connected (N/C)   |                                                                                      |

## V. Standard use condition

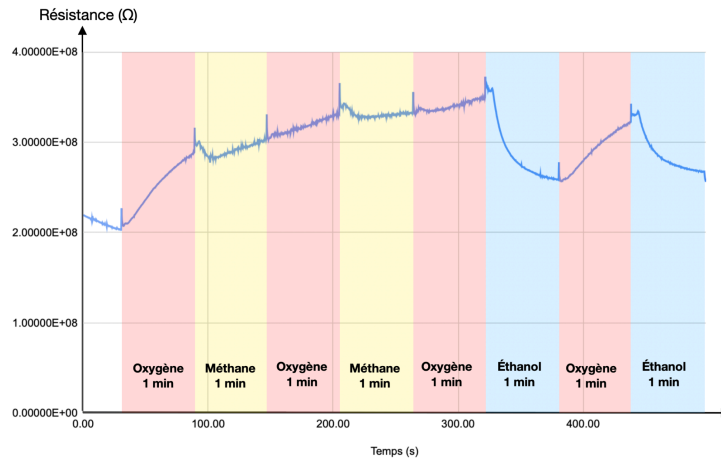
|             | Unit    | Typical Value |
|-------------|---------|---------------|
| Temperature | °C      | 20            |
| Humidity    | %       | 50            |
| Air quality | % N2/O2 | 80/20         |

## VI. Electrical characteristics

|                               | Unit | Value |         |     |
|-------------------------------|------|-------|---------|-----|
|                               |      | Min   | Typical | Max |
| Gas sensor resistance         | GΩ   | 10    | 20      | 40  |
| Temperature sensor resistance | Ω    | -     | 75      | -   |
| Heater resistance             | Ω    | -     | 90      | -   |
| Gas sensor voltage            | V    | -     | 5       | -   |
| Heater                        | V    | 10    | 12      | 20  |

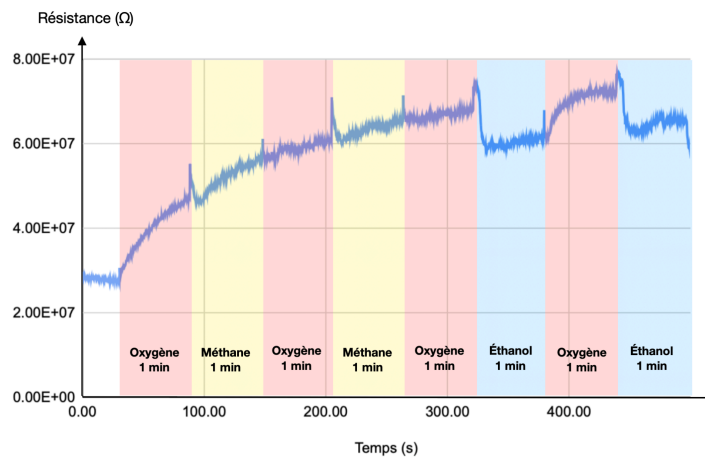
## VII. Gas sensor resistance characteristics

The sensor's resistance varies significantly in response to different gases at specific operating temperatures. Below are the typical resistance dynamics observed at 150°C, 250°C, and 350°C for O<sub>2</sub>, Ethanol (CH<sub>3</sub>CH<sub>2</sub>OH), and Methane (CH<sub>4</sub>).



*Figure 1. Resistance dynamic at 150°C*

The resistance increases and stabilises during exposure to Oxygen and Methane, with sharp peaks for Ethanol. Ethanol causes a significant resistance spike, making it easily distinguishable from Oxygen and Methane.



*Figure 2. Resistance dynamic at 250°C*

Resistance levels are lower compared to 150°C, with clearer separation between Oxygen, Methane, and Ethanol. Ethanol still causes a pronounced spike, but the baseline resistance is reduced, improving sensitivity.

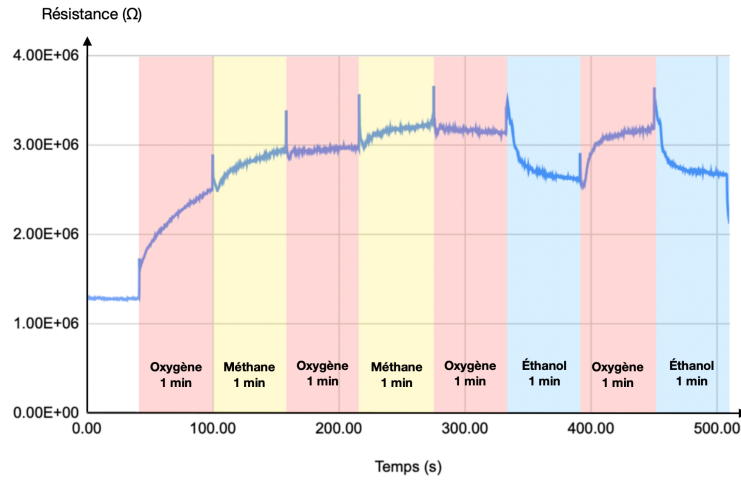


Figure 3. Resistance dynamic at 350°C

Resistance is further reduced, and the distinction between gases becomes more pronounced. Ethanol and Methane cause distinct resistance changes, while Oxygen remains stable.

## VIII. Typical application

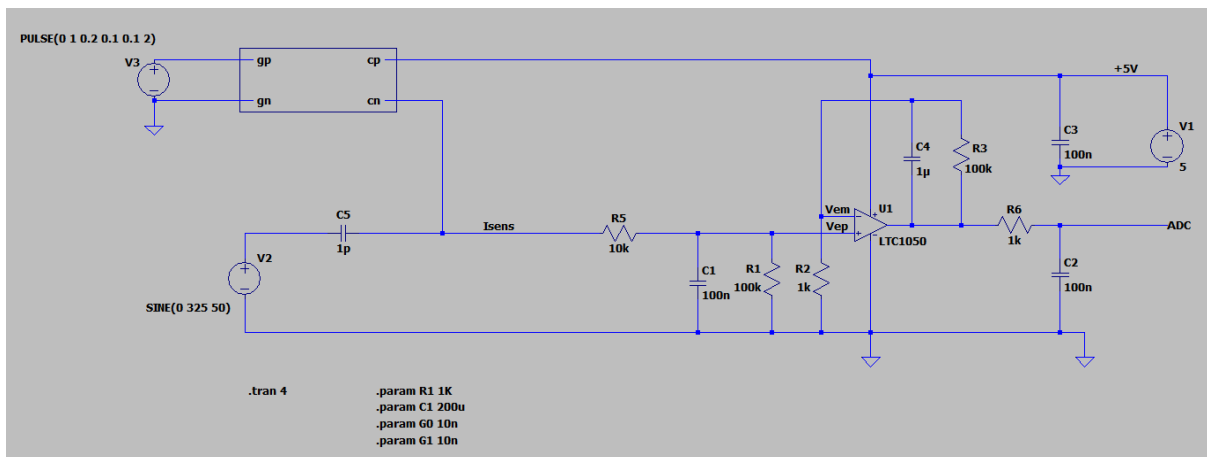


Figure 4: Example Application Circuit for  $WO_3$  Gas Sensor

To use this sensor as a smart sensor, follow the circuit in Figure 4. Higher temperatures (e.g., 350°C) improve sensitivity and reduce baseline resistance, making gas detection more accurate. The key steps are:

- Power the sensor with a PWM AC source (e.g., V1) to control the heater and minimise power consumption.
- Condition the signal using an amplifier (e.g., LTC1050) with resistors (R4, R7, R8, R9, R10) and capacitors (C5, C6, C7, C8) to filter noise.
- Connect the output to an ADC for digital processing.
- Use a microcontroller to:
  - Adjust heater temperature via PWM.
  - Read ADC values and convert them to gas concentration.
  - Trigger regeneration if the sensor saturates.

| Name  | Description                                                                     |
|-------|---------------------------------------------------------------------------------|
| V3    | PWM AC source to power and control the heater.                                  |
| V2    | AC signal source to simulate sensor input for testing.                          |
| V1    | +5V DC power supply for the amplifier and ADC.                                  |
| R1    | 100k $\Omega$ resistor, part of the amplifier feedback network.                 |
| R2    | 1k $\Omega$ resistor, part of the amplifier feedback network.                   |
| R3    | 100k $\Omega$ resistor, connected to the non-inverting input of the amplifier.  |
| R5    | 10k $\Omega$ resistor, used for current sensing (Isens).                        |
| R6    | 1k $\Omega$ resistor, part of the output stage of the amplifier.                |
| C1    | 200nF capacitor, used for noise filtering in the amplifier input stage.         |
| C2    | 100nF capacitor, used for noise filtering at the amplifier output.              |
| C3    | 100nF capacitor, used for decoupling the +5V power supply.                      |
| C4    | 1 $\mu$ F capacitor, used for stabilizing the amplifier reference voltage.      |
| C5    | 1pF capacitor, used for filtering high-frequency noise in the sensor input.     |
| U1    | LTC1050 operational amplifier, used for amplifying the sensor signal.           |
| Isens | Current sensing node for monitoring heater current.                             |
| ADC   | Analog-to-Digital Converter input, for digital processing of the sensor signal. |